AE 481 Team 01 Assignment 5

September 29, 2025

1 Engine Selection

The engines considered are the F414-GE-400, F110-PW-229, F110-PW-229A, F110-GE-129, and F110-GE-132. The selected engine is the F110-GE-132 with a maximum sea-level thrust with afterburner (A/B) of 33,093 lbs and a maximum dry (military thrust) of 17,669 lbs per engine [1]. Two of these engines will be used. The engine has the most thrust out of the potential engines with lower or similar thrust specific fuel consumption (TSFC) values.

Table 1: Engine Specifications from Nicolai [1]. F110-GE-132 has the highest thrust with lower dry TSFC and similar afterburner TSFC.

Engine	Thrust A/B (SLS-lbs)	Military Thrust (SLS-lbs)	Weight (lbs)	TSFC (Dry) (1/s)	TSFC (A/B) (1/s)
F110-GE-132	33093	17669	3990	0.68	1.90
F414-GE-400	21496	14447	2512	0.82	1.844
F110-GE-129	29474	17595	3940	0.67	1.85
F110-PW-229	29100	17800	3850	0.74	2.05
F110-PW-229A	32500	20100	3830	0.71	1.86

2 Engine Thrust Corrections

Power and airflow offtake systems can be accounted for through thrust corrections through engine installation corrections discussed by Raymer [2] and Nicolai [1]. These corrections are for pressure loss inside the inlet duct and engine bleed. Raymer states that installed engine thrust is also affected by horsepower extraction, but that this only has a very small effect on installed and thus can be ignored for initial design. If more detailed engine data is available, better estimations of installed thrust corrections can be made; however, this data is not available.

2.1 Inlet Pressure Loss Thrust Correction

The percent thrust loss is given by Raymer and Nicolai and is calculated using Eq. 1:

$$\%_{TL_{PR}} = c_r[\eta_{\text{ref}} - \eta_{\text{actual}}] * 100\% \tag{1}$$

where c_r is the inlet ram recovery correction factor and given by Eq. 2 if manufacturer's data is not available:

$$\begin{cases} c_r = 1.35 & M_\infty \leq 1 \\ c_r = 1.35 - 0.15(M_\infty - 1) & M_\infty > 1 \end{cases} \tag{2}$$

The term eta_{actual}, actual internal pressure recovery, is given by Raymer as 0.94 for an S-duct inlet.

The term eta_{ref} is dependent on the Mach number. To find an equation relating the Mach number to the actual internal pressure recovery, a regression must be performed based on numbers given by MIL-E-5008 [3]. Raymer and Nicolai give an equation based on this MIL specification, but the implementation of that equation gives an imaginary value for η_{actual} and as such, is not used. The points given by the MIL-E-5008 are given in Eq. 3:

$$\begin{cases} 0.95 & 0 < M \leq 1 \\ 0.85 & M = 2 \\ 0.60 & M = 3 \end{cases} \tag{3}$$

By fitting a quadratic in MATLAB to the points above, eta_{actual} is given by Eq. **??**

$$\begin{cases} eta_{\text{ref}} = 1 & 0 < M_\infty \leq 1 \\ eta_{\text{ref}} = -0.0750M_\infty^2 + 0.1250M_\infty + 0.9 & M_\infty > 1 \end{cases} \tag{4}$$

2.2 Engine Bleed

The percent thrust loss is given by Raymer and Nicolai and is calculated using Eq. 5:

$$\%_{TL_{EB}} = C_b \frac{\dot{m}_b}{\dot{m}_e} \tag{5}$$

where C_b is the bleed correction factor and is assumed to be 2 according to Raymer and Nicolai. The bleed mass flow to engine mass flow ratio is typically between 1 and 5%. Roskam part VI [4] gives a value of 0.04 for attack jet fighters so that is the bleed mass flow to engine mass flow ratio used for the thrust correction factor.

2.3 Installed Thrust

Installed thrust is calculated using Eq. 6.

$$T_{\text{installed}} = T(1 - \frac{\%_{TL_{PR}} + \%_{TL_{EB}}}{100}) \tag{6}$$

Both thrust corrections are assumed to be reduced separately from the given thrust, rather than by multiplying one correction by the other correction by the thrust.

It should be noted that these thrust corrections have not been implemented within the T-S plot code. These corrections will be implemented for future assignments.

3 T-S Plot

T-S plots were constructed by modifying the takeoff weight estimation code to modify empty weight by wing area and takeoff thrust. Additionally, the parasitic drag coefficient C_{D0} is modified by the planform area. The modified coefficient is inputted into the fuel fraction estimation subroutine. The general procedure is derived from algorithms in Martins [5]. Several simplifying assumptions are made to make the T-S plot process feasible. First, empty weight modifications are calculated as changes about a design thrust and wing area. Second, C_{D0} is calculated assuming a fixed $S_{\text{wet,rest}}$, calculated at the design point takeoff weight. Finally, thrust is not modified in the fuel fraction estimate. This is based on the assumption that the T-S plot is accurate near the design point. This helps prevent convergence issues, but will be revised in the future.

Conversion of T/W vs. W/S constraint lines to T-S curves is done using two methods. Either, a wing area is prescribed and the constraint lines are solved for thrust and takeoff weight, or a thrust is prescribed and the constraints are solved for wing area and takeoff weight. MATLAB's `fsolve` is used to solve the unknowns (e.g. T and W_0) as a system of equations. To make the solver converge, the unknowns are normalized by the design takeoff weight, thrust, and wing area. This helps prevent scaling issues when solving for the unknowns. Additionally, the initial solution guess is seeded using a previous solution. This helps start the solver near the new solution, when the inputs are modified by small amounts.

3.1 Objective Function Contours

Cost was chosen as the objective, as it is the primary objective discussed in the RFP, aside from maximizing mission performance. The cost prediction code from Assignment 3 was updated for this task, which primarily relies on equations from Roskam.

The specific cost used is the Airplane Estimated Cost (AEP) from Roskam. This is calculated using historical regressions for the costs from the RDTE and production phases. The primary inputs to this are the number of units and the MTOW. The number of units is 500 from the RFP, and the MTOW is taken as the maximum weight from either the strike or air-to-air mission. MTOW varies with thrust and wing area, so it is well suited for contours that depend on both variables.

3.2 Design Point and T-S Plot

The T-S plot for the given design point of 850 ft^2 and 66000 lb of thrust yields an aircraft with a takeoff weight of 74000 lb. For the air to air mission (Fig. 1), the aircraft has a dash Mach number of 1.7, a combat radius of 800 nm, and a combat time of 3 min. The feasible region is constrained by the dash, catapult launch, and conventional takeoff constraint.

For the strike mission (Fig. 2), the aircraft has a dash Mach number of 0.9 and a combat radius of 1010 nm The feasible region is constrained by the takeoff single engine rate of climb, catapult launch, and conventional takeoff constraint. Contours of aircraft unit cost are plotted in the T-S plots in millions of USD. The current design point costs roughly 119 million USD.

For future analysis, more design points need to be analyzed for lower wing areas and potentially lower thrust engines. This could result in a reduced cost while maintaining desired aircraft mission performance.

4 F/A-18E Validation

4.1 $C_{L_{\max}}$ For T-S Plot

We found that the $C_{L_{\max}}$ of the F/A-18 ranges from a low of around 1.6 from Skow [6] to a high of around 2.0 from Erickson [7]. Our $C_{L_{\max}}$ estimates fall within Roskam's ranges [8]. Roskam gives a $C_{L_{\max}}$ range of from 1.4 - 2.0 for fighter aircraft takeoff, and a range of 1.6 - 2.6 for fighter aircraft landing. Therefore, for $C_{L_{\max}}$ with half flaps and full flaps we use 1.6 and 2.0 respectively.

4.2 Weight Estimation and T-S Plot

The weight estimation code is validated using the F/A-18E mission profiles for a fighter escort and interdiction mission [9]. These are analogous to the air to air and strike mission respectively. The resulting takeoff weight estimates are compared in Table 2. The improved results is due to the new method of predicting C_{D0}, by using takeoff weight to predict wetted area. Additionally, the C_{D0} is used to predict $L/D_{\max}$ instead of using fitted plots from Raymer.

Table 2: Takeoff weight estimation is improved compared to previous estimates.

	Fighter Escort	**Interdiction**
Actual Takeoff Weight (lb)	58948	60769
Predicted Takeoff Weight (lb)	55400	59100

A T-S plot was generated for each mission and is shown in Figure 3 and 4. The F/A-18E design point was found to be within the feasible region for both. This suggests that the T-S plot code is implemented correctly. However, further review needs to be conducted, primarily with exploring other feasible design points.

5 Group Work Split

Table 3: Work Split

Person	Work Percentage	Brief Description of Work Done
Michael Chen	30%	Wrote T-S plot code, validated code with F/A-18E data, helped write report.
Charles Choi	14%	Worked on F18 CAD mock-up, parameterized so that dimensions can be substituted with actual airframe specs. Helped select engine by looking at Takeoff Thrust specification with considerations to weight, cost, and quality (corrosion and reliability.
Cristina Erskine	16%	Researched thrust installation corrections and wrote correction regression code, helped select engine, researched updated values for T-S plot code, helped write report.
James Gold	16%	Helped find installed and bleed corrections, validated $C_{L_{\max}}$ values, made corrections to the cost code, helped write report.
Santiago Ramos-Assam	12%	Helped with bleed rate and thrust data, validated $C_{L_{\max}}$ values, helped write report.
Thomas Sheridan	12%	Researched bleed air and shaft power thrust reductions. Revised cost code for more accurate AEP and COC prediction. Helped write report.

References

[1] Leland M. Nicolai. *Fundamentals of Aircraft Design*. American Institute of Aeronautics and Astronautics, 2010. ISBN: 978-1-60086-751-4.

[2] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 2018. ISBN: 9781624104909.

[3] J. Franklin Montgomery III. *The Need For Air Force Engine Stability Margin Testing For Inlet-Engine Interface Definition*. 1976.

[4] Jan Dr. Roskam. *Part 6: Preliminary Calculation of. Aerodynamic, Thrust and Power Characteristics*. Roskam Aviation and Engineering Corporation, 1987.

[5] Joaquim R.R.A. Martins. *The Metabook of Aircraft Design*. Dr. Martins, Joaquim R.R.A., 2021.

[6] Andrew Skow. *Combat Agility Management System*. Tech. rep. NASA Fourth High-Angle-Of-Attack, 1996.

[7] Gary E. Erickson. *Wind Tunnel Investigation of Vortex Flows on F/A-18 Configuration at Subsonic Through Transonic Speeds*. Tech. rep. NASA Langley Research Center, 1991.

[8] Jan Dr. Roskam. *Part 1: Preliminary Sizing of Airplanes*. Roskam Aviation and Engineering Corporation, 1985.

[9] *Standard Aircraft Characteristics F/A-18E Super Hornet*. NAVAIR00-110AF18-6. Boeing. 2001.

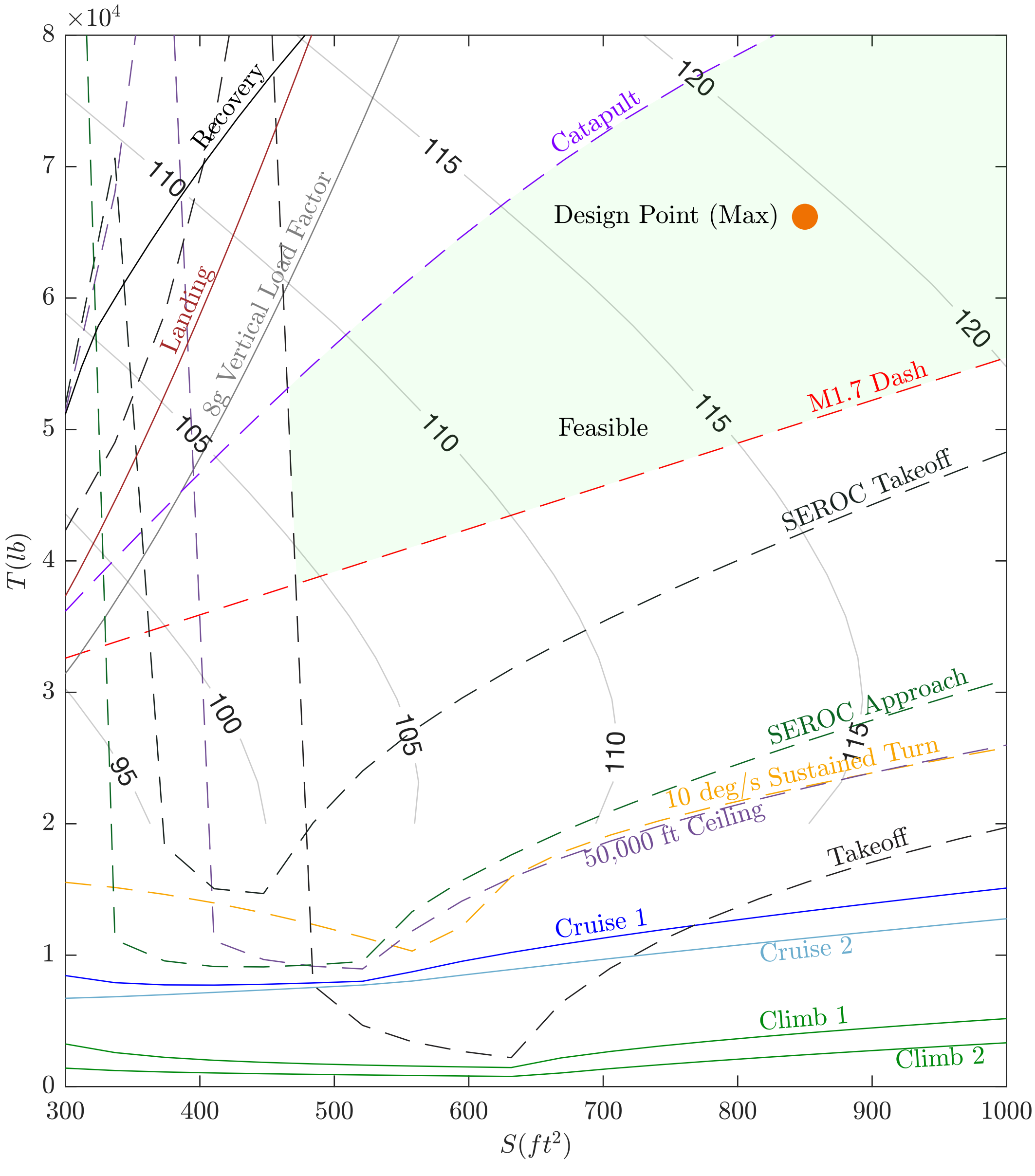

Figure 1: T-S plot with contours of cost (millions USD 2025) for the air to air mission.

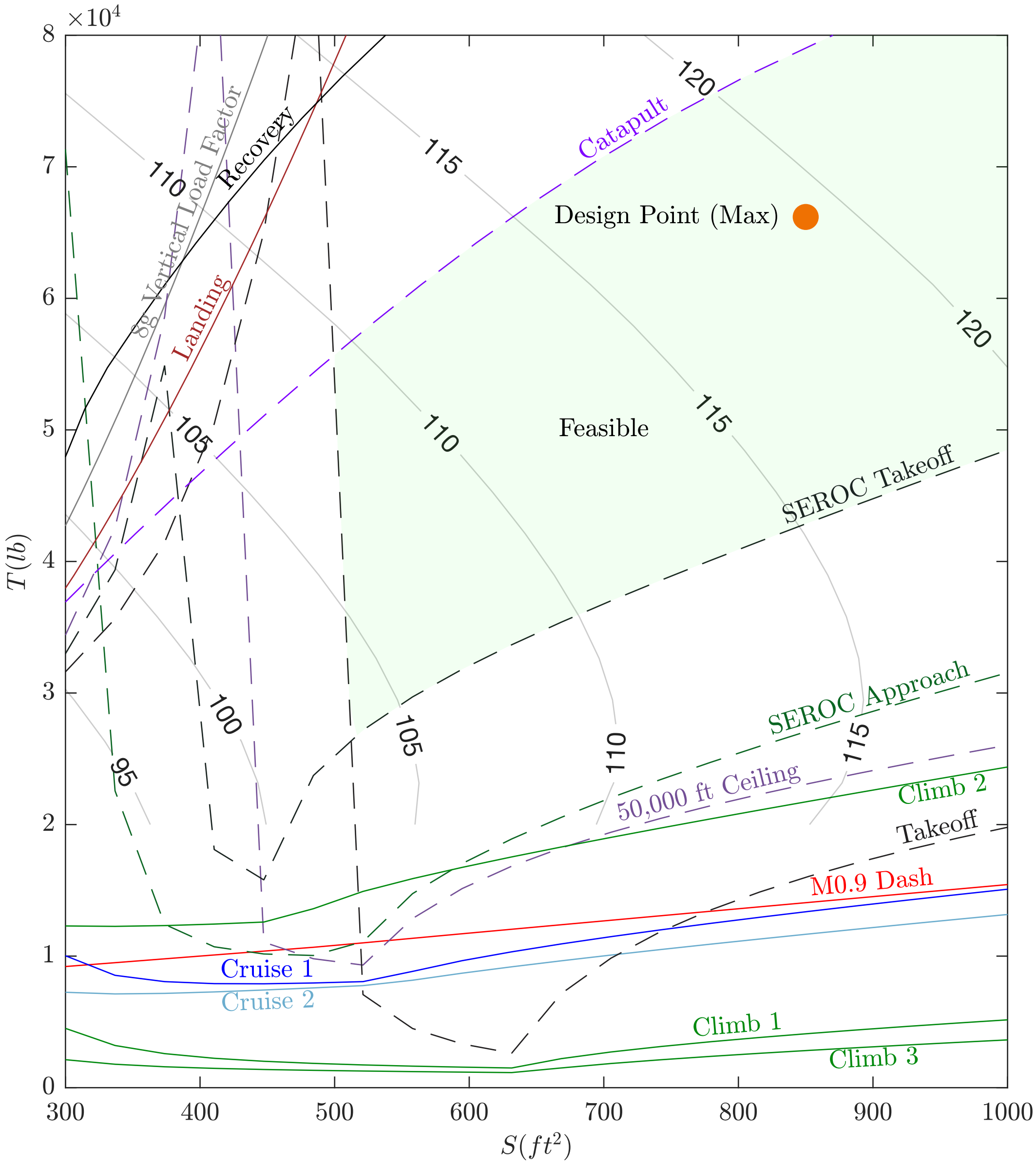

Figure 2: T-S plot with contours of cost (millions USD 2025) for the strike mission.

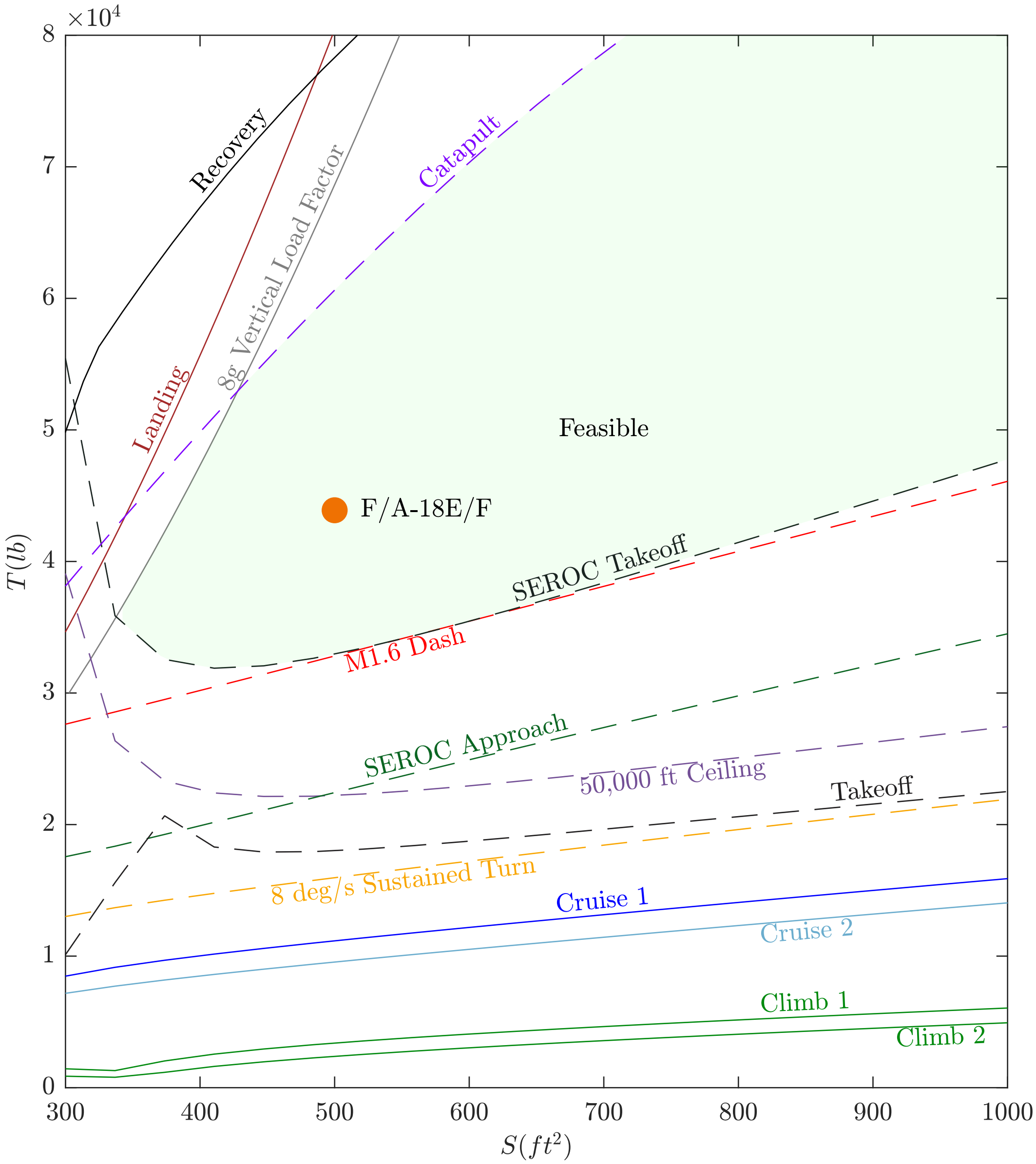

Figure 3: Validation T-S plot for the fighter escort mission for the F/A-18E.

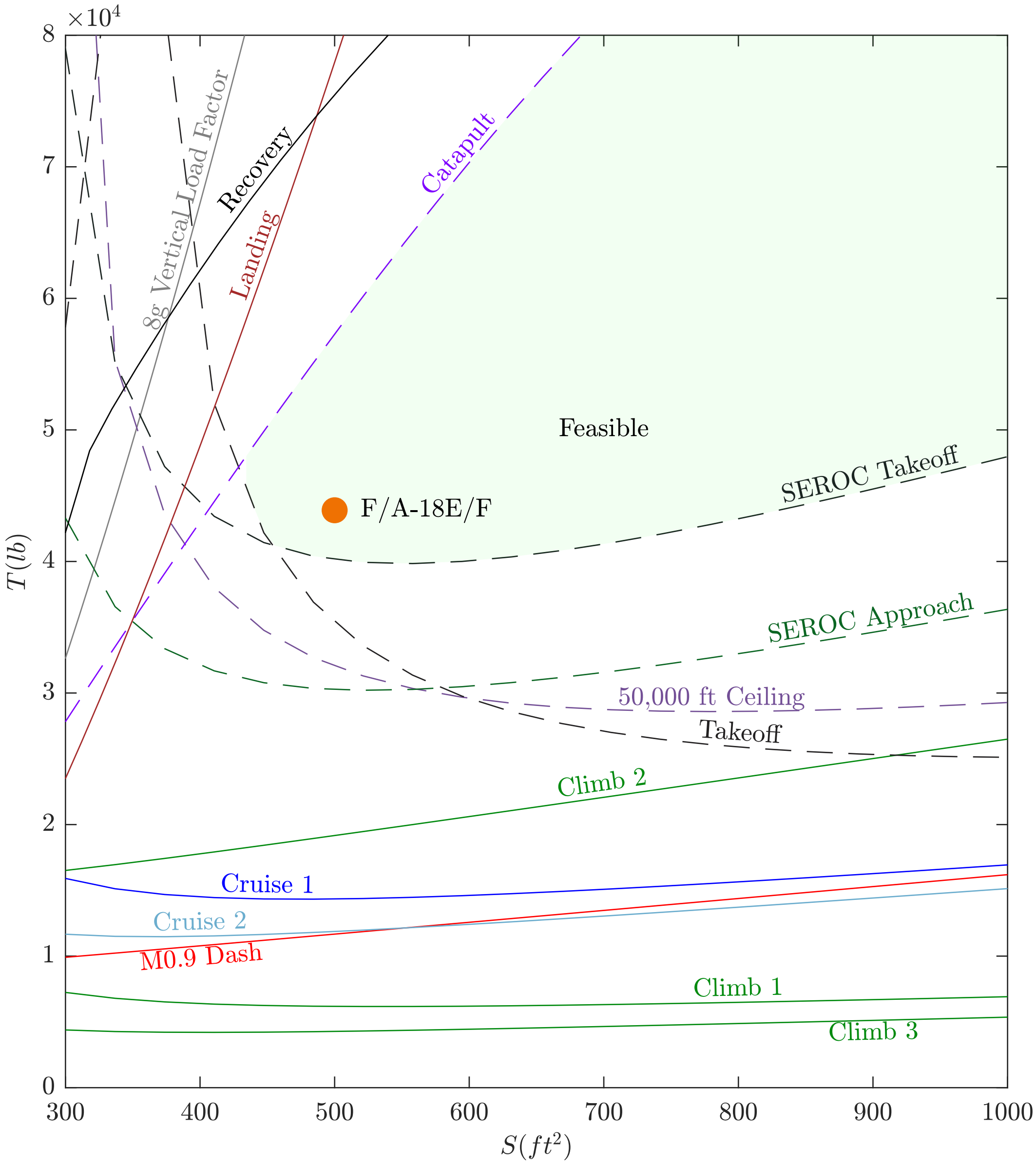

Figure 4: Validation T-S plot for the interdiction mission for the F/A-18E.